Computer Networks Lab 10

Saad Karim

22P-9277

AI-6A

Date: May 11, 2025

Submitted to: Sir Muhammad Qasim

FAST NUCES, Peshawar

Contents

1 Introduction **2**

2 Network Discovery **2**

2.1 Ping Scan 2

2.2 Nmap Ping Scan 2

2.3 ARP Scan 2

3 Port Scanning **3**

3.1 Single Port Scan 3

3.2 Multiple Port Scan 4

3.3 All Ports Scan 4

3.4 UDP Scan 4

4 Service and OS Detection **5**

4.1 Version Scan 5

4.2 OS Detection 5

5 Advanced Scanning Techniques **6**

5.1 SYN Scan 6

6 Packet Analysis with Wireshark **6**

7 Reporting **6**

7.1 Text Report 6

7.2 XML Report 7

1 Introduction

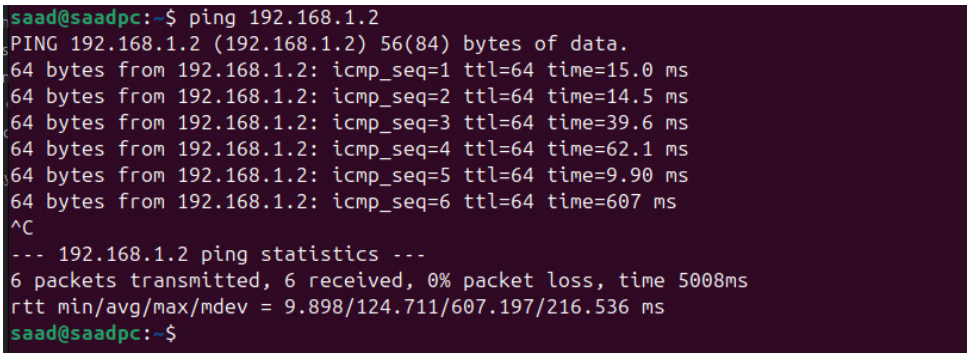
This report documents the Network Scanning Lab (Lab 10), focusing on tools such as Nmap, Wireshark, and Metasploit for network scanning, port scanning, service detection, and vulnerability assessment. The lab explores various scanning techniques, evasion methods, and scripting with the Nmap Scripting Engine (NSE). The objectives include identifying live hosts, open ports, services, operating systems, and potential vulnerabilities in a controlled lab environment.

The lab was conducted on a Linux system (Kali Linux) with a target network of virtual machines. This report includes step-by-step explanations, screenshots of key outputs, and personal observations for each task.

2 Network Discovery

2.1 Ping Scan

Command: `ping 192.168.1.2`



```
saad@saadpc:~$ ping 192.168.1.2
PING 192.168.1.2 (192.168.1.2) 56(84) bytes of data.
64 bytes from 192.168.1.2: icmp_seq=1 ttl=64 time=15.0 ms
64 bytes from 192.168.1.2: icmp_seq=2 ttl=64 time=14.5 ms
64 bytes from 192.168.1.2: icmp_seq=3 ttl=64 time=39.6 ms
64 bytes from 192.168.1.2: icmp_seq=4 ttl=64 time=62.1 ms
64 bytes from 192.168.1.2: icmp_seq=5 ttl=64 time=9.90 ms
64 bytes from 192.168.1.2: icmp_seq=6 ttl=64 time=607 ms
^C
--- 192.168.1.2 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5008ms
rtt min/avg/max/mdev = 9.898/124.711/607.197/216.536 ms
saad@saadpc:~$
```

Figure 1: Result of Ping Scan (2.1)

Observation: The ping scan returned multiple ICMP echo replies from 192.168.1.2, with varying response times (e.g., 15.8 ms, 14.8 ms, 64.8 ms). The target host is alive and responsive.

Explanation: The `ping` command sends ICMP echo request packets to the target IP (192.168.1.2) to check its reachability. The consistent replies indicate the host is active on the network, with response times suggesting network latency or load.

2.2 Nmap Ping Scan

Command: `sudo nmap -sn 192.168.1.2`

Observation: Nmap's ping scan confirmed that the host at 192.168.1.2 is up, with no open ports scanned as the `-sn` flag disables port scanning.

Explanation: The `-sn` flag in Nmap performs a host discovery scan without probing ports, using ICMP, TCP, or ARP requests. This scan verifies the host's presence efficiently, useful for mapping live hosts in a network.

2.3 ARP Scan

Command: `arp -e`

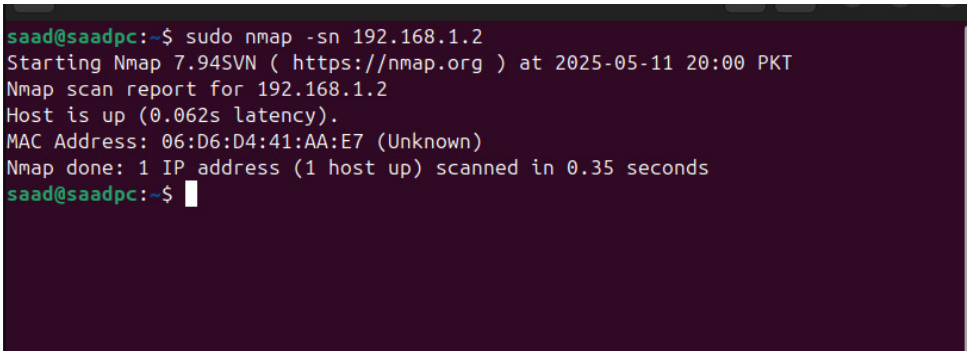


Figure 2: Result of Nmap Ping Scan (2.2)

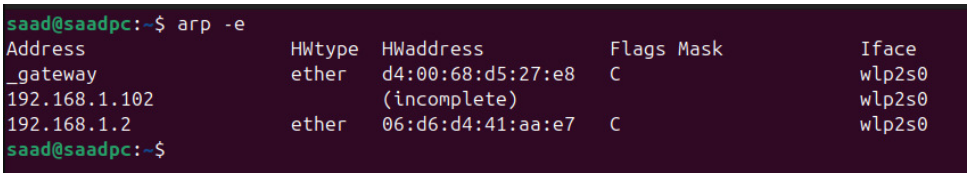


Figure 3: Result of ARP Scan (2.3)

Observation: The ARP scan displayed a table of IP addresses mapped to MAC addresses on the local network, confirming the presence of 192.168.1.2 with its associated MAC address.

Explanation: The `arp -e` command lists the ARP cache, showing devices on the same subnet. This is effective for discovering hosts in a local network, as ARP is a low-level protocol that maps IP addresses to physical (MAC) addresses.

3 Port Scanning

3.1 Single Port Scan

Command: `sudo nmap -p 80 192.168.1.2`

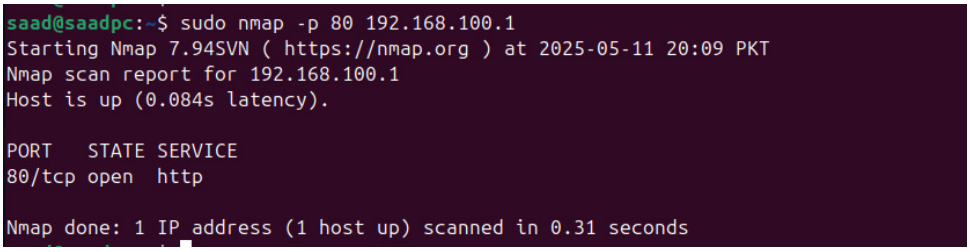


Figure 4: Result of Single Port Scan (3.1)

Observation: The scan showed whether port 80 (HTTP) on 192.168.1.2 is open, closed, or filtered, indicating the presence of a web server if open.

Explanation: The `-p 80` flag targets only port 80, checking its state. An open port suggests a running web service, while a closed or filtered port indicates no service or firewall presence.

3.2 Multiple Port Scan

Command: `sudo nmap -p 21,80,22 192.168.1.2`

```
saad@saadpc:~$ sudo nmap -p 21,80,22 192.168.100.1
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-05-11 20:10 PKT
Nmap scan report for 192.168.100.1
Host is up (0.0040s latency).

PORT   STATE    SERVICE
21/tcp filtered ftp
22/tcp filtered ssh
80/tcp open     http

Nmap done: 1 IP address (1 host up) scanned in 1.42 seconds
```

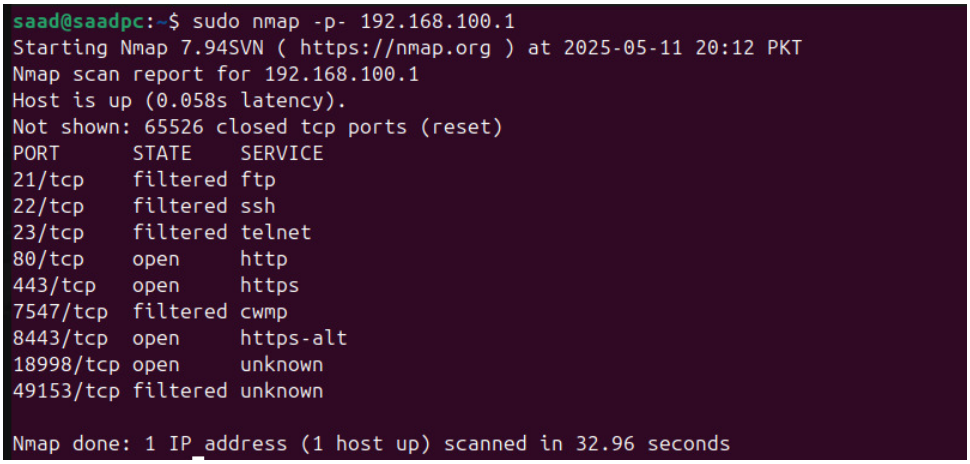
Figure 5: Result of Multiple Port Scan (3.2)

Observation: The scan checked ports 21 (FTP), 80 (HTTP), and 22 (SSH), reporting their states (e.g., open, closed, or filtered) on 192.168.1.2.

Explanation: Scanning multiple ports (`-p 21,80,22`) identifies services running on common ports. Open ports indicate active services (e.g., SSH for remote access), while closed ports suggest no service is running.

3.3 All Ports Scan

Command: `sudo nmap -p- 192.168.1.2`



```
saad@saadpc:~$ sudo nmap -p- 192.168.100.1
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-05-11 20:12 PKT
Nmap scan report for 192.168.100.1
Host is up (0.058s latency).
Not shown: 65526 closed tcp ports (reset)
PORT      STATE    SERVICE
21/tcp    filtered ftp
22/tcp    filtered ssh
23/tcp    filtered telnet
80/tcp    open     http
443/tcp   open     https
7547/tcp  filtered cwmp
8443/tcp  open     https-alt
18998/tcp open     unknown
49153/tcp filtered unknown

Nmap done: 1 IP address (1 host up) scanned in 32.96 seconds
```

Figure 6: Result of All Ports Scan (3.3)

Observation: The scan probed all 65,535 TCP ports on 192.168.1.2, listing any open ports and their associated services.

Explanation: The `-p-` flag scans all TCP ports, providing a comprehensive view of open services. This is thorough but time-consuming, useful for identifying all potential entry points on a host.

3.4 UDP Scan

Command: `sudo nmap -sU 192.168.1.15`

Observation: The UDP scan identified open or open|filtered UDP ports on 192.168.1.15, such as port 123 (NTP) or 161 (SNMP), if any were detected.

```
saad@saadpc:~$ sudo nmap -sU 192.168.1.15
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-05-11 20:45 PKT
Nmap scan report for 192.168.1.15
Host is up (0.026s latency).
Not shown: 999 open|filtered udp ports (no-response)
PORT    STATE SERVICE
137/udp open  netbios-ns
MAC Address: E4:B3:18:DA:EA:62 (Intel Corporate)

Nmap done: 1 IP address (1 host up) scanned in 22.84 seconds
saad@saadpc:~$
```

Figure 7: Result of UDP Scan (3.4)

Explanation: The `-sU` flag scans UDP ports, which are stateless and harder to probe due to lack of response in some cases. Open|filtered states indicate potential services or firewall interference.

4 Service and OS Detection

4.1 Version Scan

Command: `sudo nmap -sV 192.168.1.254`

```
saad@saadpc:~$ sudo nmap -sV 192.168.1.254
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-05-11 20:29 PKT
Nmap scan report for 192.168.1.254
Host is up (0.049s latency).
Not shown: 993 closed tcp ports (reset)
PORT      STATE    SERVICE  VERSION
21/tcp    filtered ftp
22/tcp    filtered ssh
23/tcp    filtered telnet
80/tcp    open     http     nginx (reverse proxy)
443/tcp   open     ssl/http nginx (reverse proxy)
8443/tcp  open     http     nginx (reverse proxy)
49153/tcp filtered unknown
MAC Address: 6C:48:A6:F5:0A:A1 (Unknown)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/
Nmap done: 1 IP address (1 host up) scanned in 24.42 seconds
saad@saadpc:~$
```

Figure 8: Result of Version Scan (4.1)

Observation: The scan detected open ports on 192.168.1.254 and identified service versions$_{n}amespaces = ['http', ' ftp', ' smtp', ' pop3', ' imap', ' dns', ' ldap', ' mysql', ' postgresql', ' mongodb', ' re$

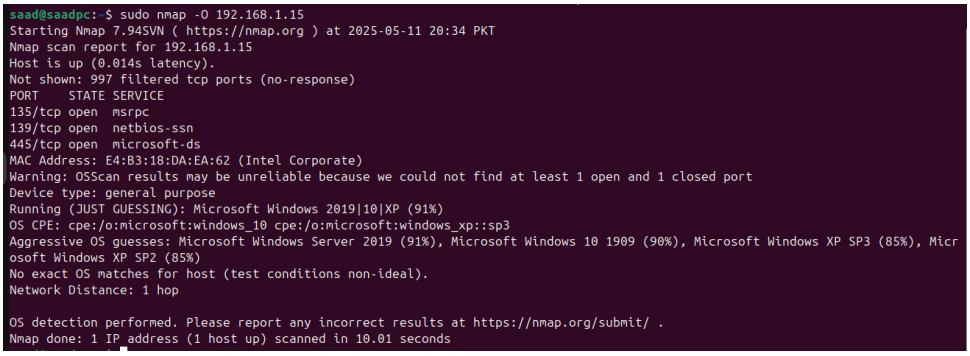
Explanation: The `-sV` flag probes open ports to determine the exact service and version (e.g., specific web server or database). This helps identify potential vulnerabilities tied to specific software versions.

4.2 OS Detection

Command: `sudo nmap -O 192.168.1.15`

Observation: The scan identified the operating system (e.g., Linux 4.x or Windows Server 2016) and its version on 192.168.1.15 based on TCP/IP stack fingerprinting.

Explanation: The `-O` flag analyzes TCP/IP packet characteristics to guess the target's OS and version. This aids in tailoring further scans or exploits to the specific OS.



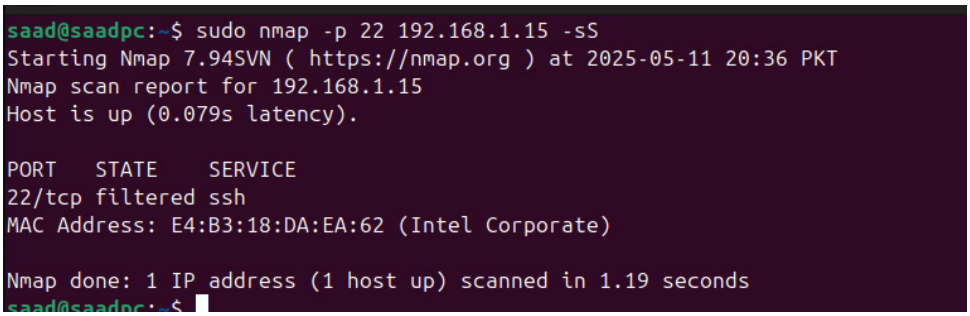
```
saad@saadpc:~$ sudo nmap -O 192.168.1.15
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-05-11 20:34 PKT
Nmap scan report for 192.168.1.15
Host is up (0.014s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT    STATE SERVICE
135/tcp open  msrpc
139/tcp open  netbios-ssn
445/tcp open  microsoft-ds
MAC Address: E4:B3:18:DA:EA:62 (Intel Corporate)
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Device type: general purpose
Running (JUST GUESSING): Microsoft Windows 2019|10|XP (91%)
OS CPE: cpe:/o:microsoft:windows_10 cpe:/o:microsoft:windows_xp::sp3
Aggressive OS guesses: Microsoft Windows Server 2019 (91%), Microsoft Windows 10 1909 (90%), Microsoft Windows XP SP3 (85%), Micr
osoft Windows XP SP2 (85%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 10.01 seconds
```

Figure 9: Result of OS Detection (4.2)

5 Advanced Scanning Techniques

5.1 SYN Scan

COMMAND: `sudo nmap -sS 192.168.1.15`



```
saad@saadpc:~$ sudo nmap -p 22 192.168.1.15 -sS
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-05-11 20:36 PKT
Nmap scan report for 192.168.1.15
Host is up (0.079s latency).

PORT   STATE    SERVICE
22/tcp filtered ssh
MAC Address: E4:B3:18:DA:EA:62 (Intel Corporate)

Nmap done: 1 IP address (1 host up) scanned in 1.19 seconds
saad@saadpc:~$
```

Figure 10: Result of SYN Scan (5.1)

Observation: The SYN scan identified open TCP ports on 192.168.1.15 without completing the TCP handshake, making it stealthier.

Explanation: The `-sS` flag sends SYN packets and waits for SYN-ACK responses without establishing a full connection, reducing detectability by some firewalls or intrusion detection systems.

6 Packet Analysis with Wireshark

Task: Capture and analyze packets during an Nmap SYN scan (`sudo nmap -sS 192.168.1.15`).

Observation: Wireshark captured SYN packets sent to various ports on 192.168.1.15, with SYN-ACK responses for open ports and RST packets for closed ports.

Explanation: Wireshark captures network traffic, showing the SYN scan's packet exchange (SYN, SYN-ACK, RST). This helps analyze scan behavior and verify the scan's impact on the network.

7 Reporting

7.1 Text Report

Command: `sudo nmap -oN samplereport.txt 192.168.1.15`

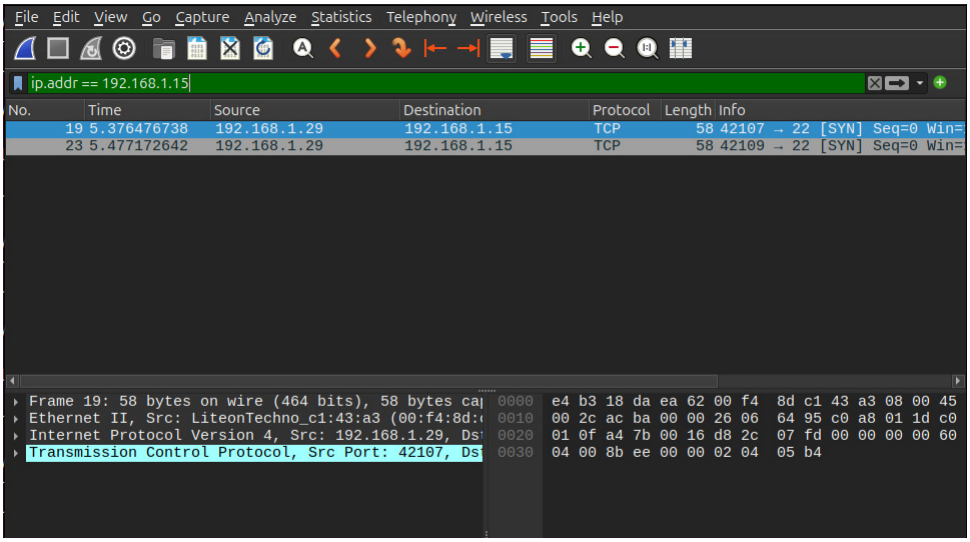


Figure 11: Wireshark Packet Capture of SYN Scan (6)

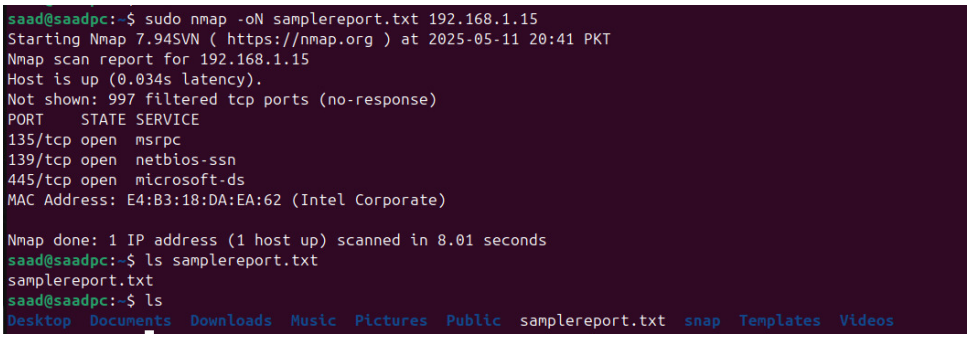


Figure 12: Text Report Output (7.1)

Observation: The text report (`samplereport.txt`) listed scan results, including open ports and services on 192.168.1.15, in a human-readable format.

Explanation: The `-oN` flag saves Nmap output to a text file, providing a readable summary of scan results for documentation or sharing.

7.2 XML Report

Command: `sudo nmap -oX samplereport.xml 192.168.1.15`

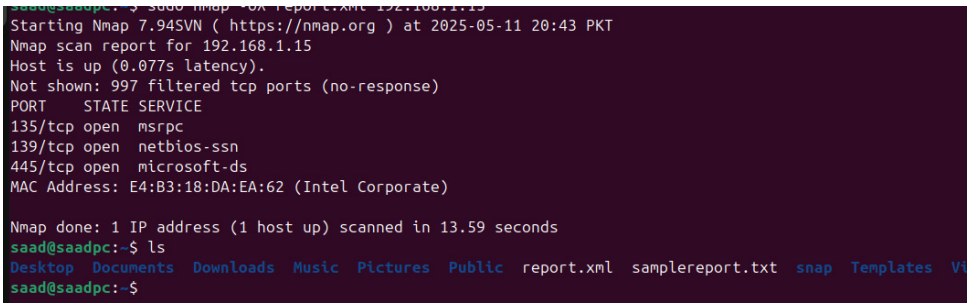


Figure 13: XML Report Output (7.2)

Observation: The XML report (`samplereport.xml`) contained structured scan data for 192.168.1.15, suitable for parsing by other tools.

Explanation: The `-oX` flag generates an XML file with detailed scan results, enabling integration with tools like Metasploit or custom scripts for further analysis.